

2.4.3 Use Flash Loader Demonstrator to burn the firmware

Flash Loader Demonstrator is an ISP download software officially released by ST. It can be used to program STM32 or STM8 chips via the UART interface. It supports burning files in *.s19, *.hex, and *.bin formats. Now let's introduce the method of using it to program STM8 microcontrollers.

First, connect the STM8 development board to the computer via the UART interface, install the Flash Loader Demonstrator software, and open it after installation. The main interface is shown in Figure 2.54. First, select the UART port number of the computer connected to the STM8 development board in the "Port Name" drop-down menu; then, in the "Echo" drop-down menu, select the "Echo Mode" option (this option is different for STM8S and STM8L).

Then, reset the STM8 microcontroller development board and immediately click the "Next" button. If the interval is too long, the data sent by the STM8 chip BootLoader will be missed, and the Flash Loader Demonstrator will fail to connect to the STM8. When the Flash Loader Demonstrator successfully connects to the STM8, it will jump to the interface shown in Figure 2.55, which is used to select the STM8 device type. The device type is classified based on the on-chip FLASH capacity of the STM8 chip. The main chip used in our development board is STM8S105C6, with an internal FLASH capacity of 32K. Therefore, select the "STM8_32K" option in the "Target" drop-down menu.

Then, click the "Next" button, and the software interface will switch to the one shown in Figure 2.56. This interface is used to select the file to be programmed. In the "Download from file" selection box, locate the file path to the file that needs to be programmed. Click the "Next" button, and the selected file will be programmed into the on-chip FLASH of the STM8 microcontroller. The software interface after a successful programming is shown in Figure 2.57.

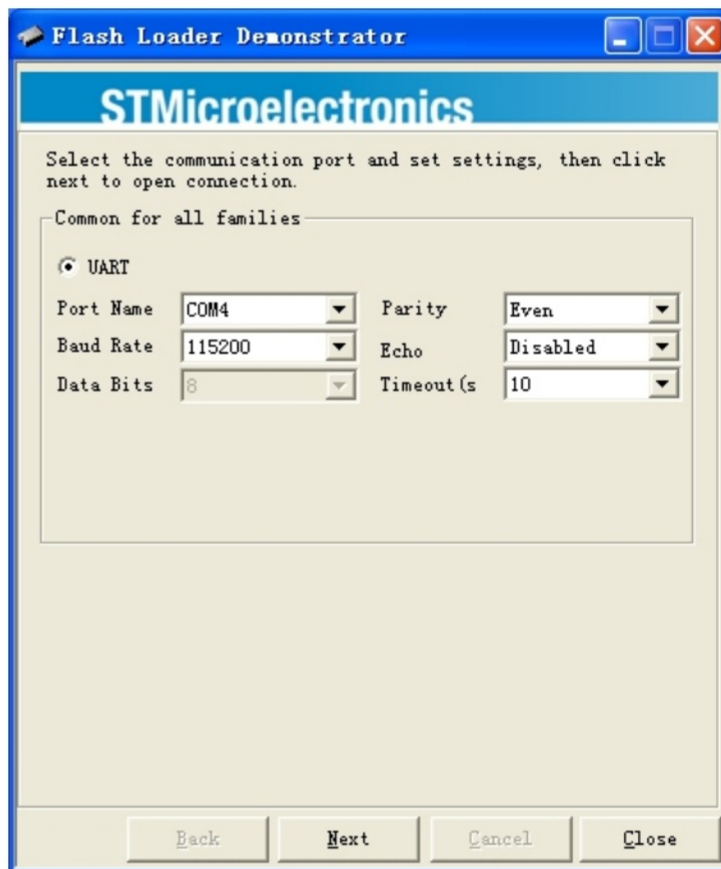


Figure 2.54 Main interface of the Flash Loader Demonstrator software

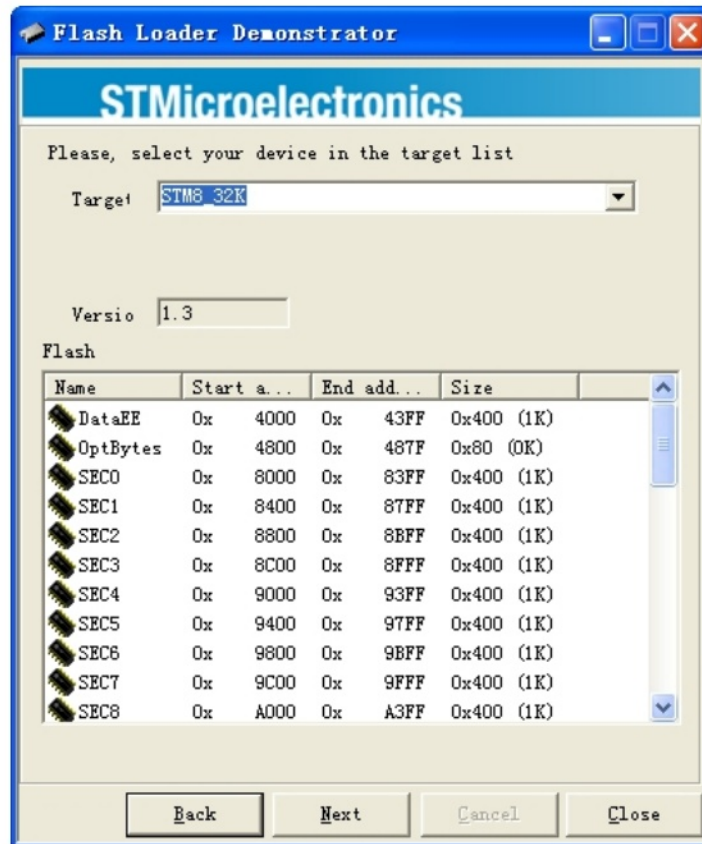


Figure 2.55 Chip Device Type Selection Interface

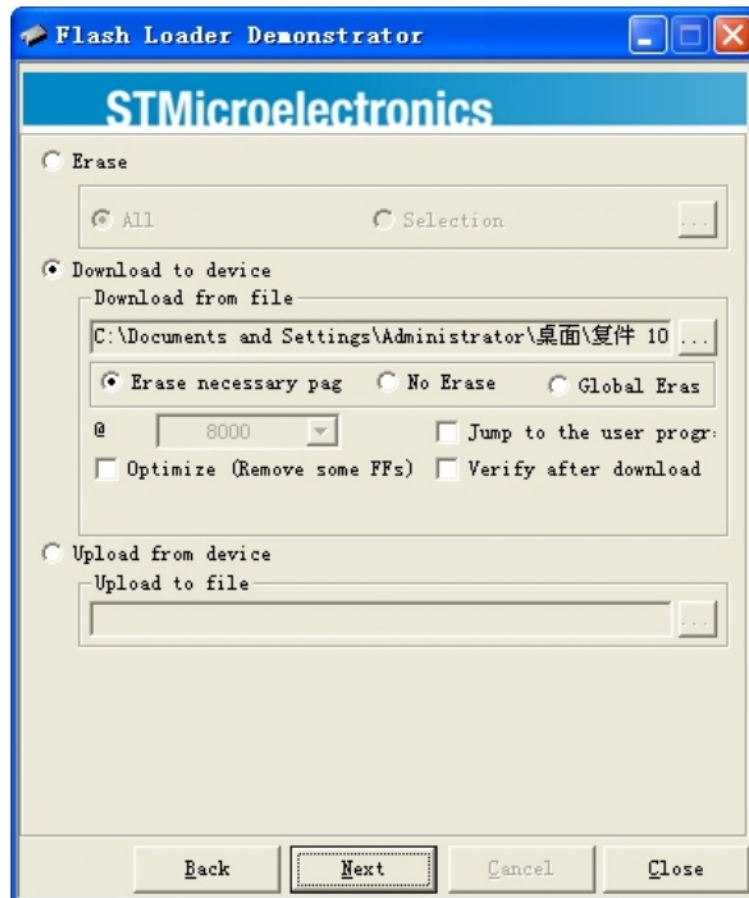


Figure 2.56 File Selection Interface for Burning

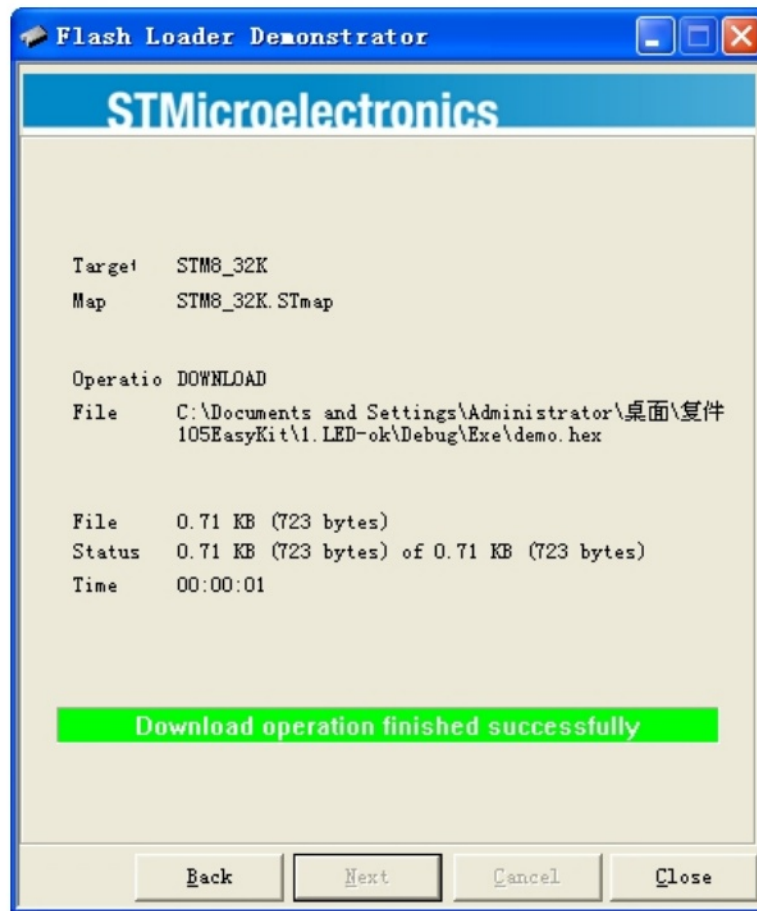


Figure 2.57 Interface for successful program burning

Finally, click the "Close" button to close the Flash Loader Demonstrator software. Then reset the STM8 development board. At this point, you will find that the STM8 is running according to the process predetermined in the program file. The Flash Loader Demonstrator only copies the program file to the FLASH of the STM8 microcontroller. After a reset, the pointer jumps to the user program for execution. If you want the program to run automatically after being burned, you need to check the "Jump to the user program" option in the interface shown in Figure 2.56.

